Free and Open Source Tools for Research

Jesse Oldroyd

June 15, 2026

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What is FOSS?

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- that is freely available (usually under a permissive license);

- has accessible source code.

Free vs. Open

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Philosophy of Free Software

The philosophy of free software can be encapsulated by *four essential freedoms* [2]:

1. to use the software.
2. to inspect and change the software.
3. to share copies of the original software.
4. to share copies of the modified software.

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2. Free software is often text-based. Since so many use this common format, it can be relatively painless to chain multiple tools together into a single workflow.
3. Given how permissive many open source tools are, it's possible (and often expected) for you to customize the tool to your own needs.

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2. FOSS tools are also more subject to becoming "abandonware" as maintaining them is time-consuming—and often thankless—work.
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Types of Tools

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- One of the most fundamental tools is a text editor.

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Tools for Computation

- R
- Python with pandas, Seaborn and SciPy
- Python with NumPy
- Python with SymPy

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Python Code for Statistics

```python
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
sns.set_theme(style="dark")

# Simulate data from a bivariate Gaussian
n = 10000
mean = [0, 0]
cov = [(2, .4), (.4, .2)]
rng = np.random.RandomState(0)
x, y = rng.multivariate_normal(mean, cov, n).T

# Draw a combo histogram and scatterplot with density contours
f, ax = plt.subplots(figsize=(6, 6))
sns.scatterplot(x=x, y=y, s=5, color=".15")
sns.histplot(x=x, y=y, bins=50, pthresh=.1, cmap="mako")
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Listing 1: Example Python code for statistical visualization

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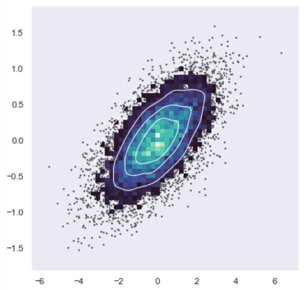
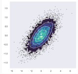


Figure 1: Data visualization produced by 1



IDE vs. Text Editor

- When using programming languages, you will need a program to interact with the language.

- There is often a choice between an *integrated development environment* (IDE) and a text editor.

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- git
- harper-ls and LanguageTool

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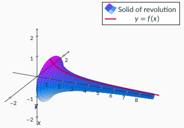
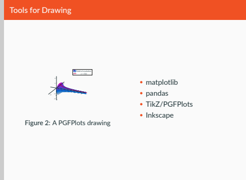


Figure 2: A PGFPlots drawing

- matplotlib
- pandas
- TikZ/PGFPlots
- Inkscape



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- The tools mentioned so far are not mutually exclusive!

- These tools have specific purposes, but can be chained together to form a powerful workflow.

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To walk you through the process of choosing a license, you can use tools such as https://choosealicense.com/.

1. GitHub will actually prompt you to provide a license.
2. In particular, can it be used in proprietary software?

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- how your work can be modified and shared.
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- Despite their utility, these tools often take longer to learn and are not as immediately intuitive as their proprietary counterparts.

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References

[1] *Open Source - GNU Project - Free Software Foundation (FSF)*. July 3, 2007. URL: https://web.archive.org/web/20070703091340/https://www.gnu.org.ua/philosophy/free-software-for-freedom.html (visited on 06/14/2026).

[2] *Philosophy of the GNU Project - GNU Project - Free Software Foundation*. URL: https://www.gnu.org/philosophy/philosophy.html (visited on 06/14/2026).

[3] *Choose a License – Open Source Programs Office*. URL: https://ospo.library.jhu.edu/learn-grow/licensing-overview/choose-a-license/ (visited on 06/14/2026).

References

[1] Open Source - GNU Project - Free Software Foundation (FSF). July 3, 2007. URL: https://web.archive.org/web/20070703091340/https://www.gnu.org.ua/philosophy/free-software-for-freedom.html (visited on 06/14/2026).

[2] Philosophy of the GNU Project - GNU Project - Free Software Foundation. URL: https://www.gnu.org/philosophy/philosophy.html (visited on 06/14/2026).

[3] Choose a License – Open Source Programs Office. URL: https://ospo.library.jhu.edu/learn-grow/licensing-overview/choose-a-license/ (visited on 06/14/2026).